Improving DDoS Attack Timeline Reconstruction Accuracy through Multi-Source Forensic Correlation

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Introduction:

Distributed Denial of Service (DDoS) attacks threaten modern networked systems by disrupting service availability and overloading resources. While mitigation techniques are widely studied, less attention has been given to accurate post attack forensic reconstruction. This study evaluates whether integrating multiple forensic data sources improves the precision of DDoS attack timeline reconstruction.

A controlled virtual testbed with attacker, victim, and analyst machines was developed using Wireshark, Zeek, ELK Stack, and Grafana. Controlled SYN flood simulations were performed, and ground truth timestamps were recorded for accuracy evaluation. Results show that multi source correlation reduces temporal error and improves identification of attack onset and recovery phases. These findings highlight the value of integrated forensic workflows for enhancing analytical precision in cybersecurity research and practice.

Problem Statement:

Single-source forensic analysis suffers from limited visibility, temporal inconsistencies, and lack of cross-source validation, leading to unreliable and incomplete reconstruction of cyber incidents.

Research Questions

RQ1: The accuracy of reconstructing a DDoS attack timeline using an integrated pipeline of Wireshark, Zeek, ELK, and Grafana.
RQ2: The extent to which temporal alignment and cross-host correlation improve reconstruction accuracy compared to single-source analysis, reflecting limitations identified in earlier studies .
RQ3: How visualization through dashboards influences analysts' understanding of attack severity, progression, and forensic significance, consistent with findings that visual analytics reveal patterns not easily observed in raw logs .

Objective: Compare single-source and multi-source forensic analysis to improve DDoS timeline reconstruction accuracy.

Methodology

A controlled experimental design was used to compare single source and multi source forensic analysis during simulated DDoS attacks.

Experimental Testbed

A virtual environment was created with three machines:

- Attacker (Kali Linux)
- Victim System
- Analyst (Ubuntu)

Controlled SYN flood attacks were performed with recorded ground truth timestamps to ensure repeatability and accurate evaluation.

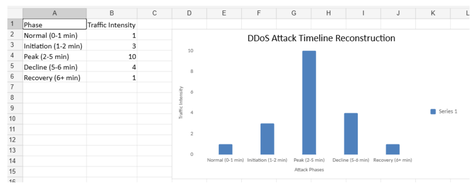
Data Collection Tools

- Wireshark for packet capture
- Zeek for network log generation
- ELK Stack for centralized log analysis
- Grafana for timeline visualization

Evaluation Metric

Mean Timestamp Deviation (ms) was used to measure reconstruction accuracy by comparing reconstructed attack timelines with actual ground truth timestamps.
Lower timestamp deviation indicates higher reconstruction accuracy.

Attack Timeline Reconstruction



Column chart showing DDoS attack progression across normal, initiation, peak, decline, and return to normal phases. It validates attack start time, peak activity, and attack end using ground truth timestamps and multi-source correlation from Wireshark, Zeek, and Grafana.

Results

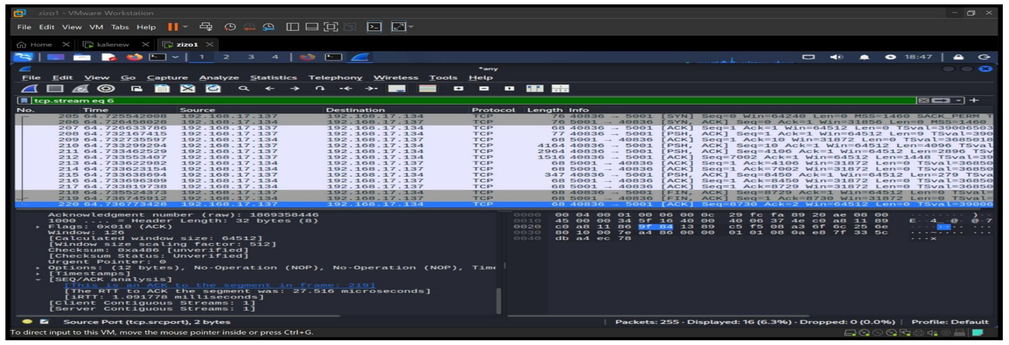
Quantitative Analysis

Multi source correlation reduced mean timestamp deviation compared to single source analysis
Improved detection of attack start time, peak traffic, and recovery phase
Lower temporal error was observed across repeated attack simulations

Qualitative Analysis

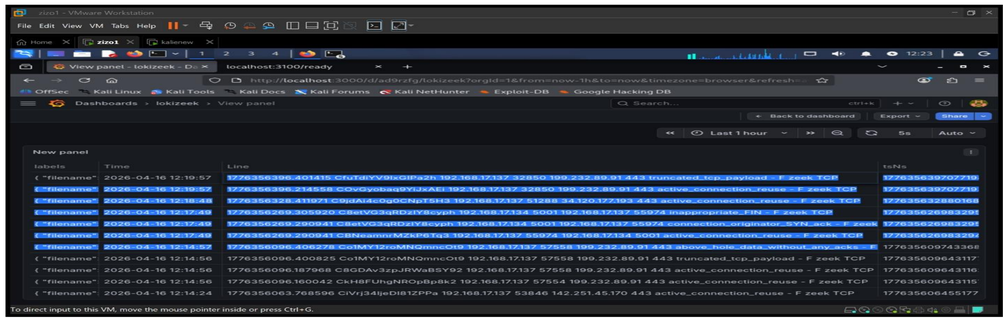
Cross source validation improved confidence in forensic reconstruction
Zeek logs and Wireshark packet captures confirmed abnormal SYN flood behavior
Grafana dashboards clearly visualized attack progression and system impact
Better understanding of attack severity and timeline reconstruction was achieved

Wireshark Packet Capture



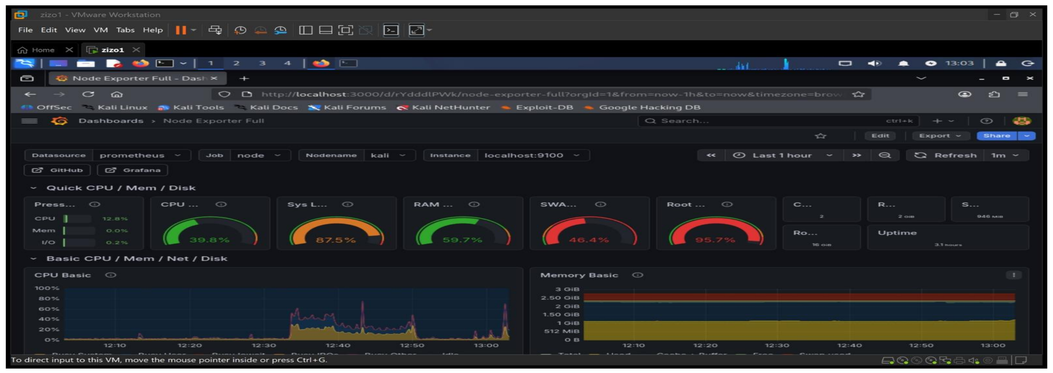
Wireshark packet capture showing repeated TCP connections between source IP 192.168.17.137 and destination IP 192.168.17.134 on port 5001. SYN, SYN-ACK, ACK, and FIN packets confirm client-server handshake behavior and abnormal traffic patterns during the DDoS attack.

Zeek Connection Logs



Zeek logs showing high-frequency TCP connections, incomplete handshakes, and abnormal states such as truncated payloads and connection reuse, providing forensic evidence of DDoS attack activity.

Grafana Dashboard



Grafana dashboard showing abnormal spikes in CPU usage, system load, and network activity, providing forensic evidence of attack progression, system stress, and DDoS impact.

Conclusion and Future Work

Multi source forensic correlation significantly improves the accuracy and reliability of DDoS attack timeline reconstruction. By combining packet level evidence, structured logs, and visualization dashboards, the system reduces temporal inconsistencies and strengthens forensic analysis. This approach supports faster incident response and more reliable cybersecurity investigations.

Future work will focus on integrating real time intrusion detection systems, applying machine learning based anomaly detection, and expanding the testbed to larger scale distributed environments to improve forensic accuracy and real-world applicability.

References: 1. Vern Paxson, "Bro: A System for Detecting Network Intruders in Real Time," *Computer Networks*, 1999.
2. National Institute of Standards and Technology, *Computer Security Incident Handling Guide (SP 800-61 Rev. 2)*, 2012.